

Week 1 — Image & Video Basics

Python-OpenCV setup, image I/O (imread, imshow, imwrite), image as numpy array, color spaces (BGR/RGB/HSV/Gray)

Basic ops — resizing, cropping, rotation, flipping, ROI slicing

Drawing shapes/text, basic arithmetic on images (blending, bitwise ops)

Mini task — build a script that reads a folder of images, converts to grayscale, resizes, saves output batch

Week 2 — Filtering, Thresholding, Morphology

Convolution concept, kernels, blurring (Gaussian, median, bilateral)

Thresholding (simple, adaptive, Otsu)

Morphological ops (erosion, dilation, opening, closing)

Task: Take a noisy scanned document image → clean it up using blur + threshold + morphology

Week 3 — Contours, Shape & Object Analysis

Contour detection & hierarchy, bounding boxes, min enclosing circle

Shape analysis: counting objects, filtering by area/aspect ratio

Task (both): Given an image of scattered objects (coins, screws, or similar — good analogy to your CNC parts), write a script to count and classify by size/shape

Week 4 — Features, Matching & Intro to Detection Models

Template matching, homography basics (panorama or alignment mini demo)

Intro to Haar cascades → face/eye detection (classic, good stepping stone)

Bridge to Month 2: brief intro to YOLO/pretrained object detection models — just running inference on a pretrained model, not training yet